

CODE SPORT



In this month's column, we discuss a few computer science interview questions.

As is our practice, we start the New Year with a medley of computer science interview questions on algorithms, data structures, machine learning and NLP. Before we do that, I wanted to point our readers to a set of useful sites that discuss computer science interview questions:

- <https://github.com/jwasham/coding-interview-university>
- <https://yangshun.github.io/tech-interview-handbook/>
- <https://github.com/Olshansk/interview>
- <https://github.com/donnemartin/system-design-primer>
- <https://cheatsheet.dennyzhang.com/cheatsheet-leetcode-a4>

These are pretty useful for preparing for coding interviews. Now let us get started on our questions for this month's column.

- (1) When we input a query into a search engine such as Google, we get an 'auto complete' suggestion for it based on past queries that have been tried on the search engine. Can you explain how you would design an auto-complete mechanism that will scale well across the millions of queries that can happen concurrently at a time? What is the data structure that you will use, which can store past queries and automatically provide suggestions? Would you store the complete query as is or would you break it into its constituent words? Given that scalability and correctness is key for this problem, can you explain how the choice of your data structure will address both?
- (2) For Covid-19, a diagnostic test has been developed. The new test is 81 per cent reliable (i.e., it misses 19 per cent of cases where the patient actually has the disease). It can wrongly label a patient as having Covid in 10 per cent of the cases when the patient actually does not have the disease. You are told nothing about the patient's symptoms or any other clinical information, except that the patient has tested positive for the Covid diagnostic test. Can you

figure out whether the patient actually has Covid-19 from the above given information? If yes, how?

If not, why not? Now you are given the additional information that in the patient's city currently, Covid-19 is present in only 1 per cent of the total population. Can you now compute the probability of whether the patient actually has Covid or not?

- (3) Given that deep neural networks need large amounts of training data to generalise well, can you explain methods by which you can artificially increase the size of your training data? (Clue: Think of transformations such as scaling, rotation and translation. While it may be obvious in case of image data sets to employ these transformations to increase the size of training data, what kind of transformations can you employ in case of text data?)
- (4) Regularisation techniques are widely used to avoid overfitting of the neural network parameters to training data so that the constructed network can still generalise well for unseen test data. Can you explain the differences between L1 regularisation and L2 regularisation methods? When would you use L1 regularisation as compared to L2 regularisation? If you change the network learning rate, does it impact the choice of your regularisation parameter lambda?
- (5) Most of you must be familiar with dropout techniques in neural networks. How is dropout helpful in avoiding overfitting? How does one apply a dropout technique in case of recurrent neural networks?
- (6) In information retrieval, you are given a set of documents and a query. You are asked to retrieve the set of documents that are most relevant to the query. Many of the common information retrieval techniques are built around the measure of TF-IDF, where TF stands for 'term frequency', i.e., the number of times a query keyword appears in a document. For example, given the query: 'Virat Kohli's one day match appearances', I am asked to find all documents relevant to this query. It is obvious that